Pharmacotherapy 2: Ultimate Question Bank

Batch 3: Lecture 3 (Migraine & Opioid Analgesics)

50 High-Yield Exam Questions with Clinical Explanations

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51. According to the vascular theory, Phase 1 of a Migraine attack (the Aura phase) involves:

A) Vasodilation

B) Vasoconstriction

C) Neurogenic inflammation

D) Platelet aggregation

> 💡 **Hint:** The Aura phase (like visual blur as the Dr. mentioned) is characterized by cranial vasoconstriction, followed by painful vasodilation in Phase 2.

52. Phase 2 of a Migraine attack is characterized by:

A) Vasoconstriction

B) Vasodilation of cranial vessels causing pulsating pain

C) Nerve compression

D) Muscle spasm

> 💡 **Hint:** The throbbing/pulsating headache is due to the sequential massive vasodilation of cranial blood vessels.

53. Triptans (e.g., Sumatriptan) act as agonists at which specific serotonin receptors?

A) 5-HT2A

B) 5-HT1B and 5-HT1D

C) Dopamine D2

D) Alpha-adrenergic

> 💡 **Hint:** Sumatriptan stimulates 5-HT1D receptors causing targeted vasoconstriction in the cranial blood vessels.

54. The mechanism of Triptans includes vasoconstriction and the inhibition of:

A) Histamine release

B) Release of pain neurotransmitters from the trigeminal nerve

C) Prostaglandin synthesis

D) Acetylcholine release

> 💡 **Hint:** Triptans are highly specific; they constrict the dilated vessels and stop the trigeminal nerve from releasing inflammatory pain peptides.

55. Triptans are best used for migraines that are:

A) Mild and infrequent

B) Moderate to severe, especially with Aura

C) Tension-type headaches

D) Cluster headaches only

> 💡 **Hint:** For mild migraines, simple analgesics (Aspirin/Acetaminophen) are sufficient. Triptans are the abortive therapy of choice for moderate-severe attacks.

56. Ergotamines are generally less preferred than Triptans for acute migraines due to:

A) Higher cost

B) Lower efficacy, more side effects, and severe vascular toxicity

C) Lack of vasoconstrictor effect

D) Extremely short half-life

> 💡 **Hint:** Ergotamine is a non-selective vasoconstrictor that can cause severe systemic ischemia and has a high side effect profile.

57. First-line prophylactic medication for continuous Migraine prevention is:

A) Sumatriptan

B) Propranolol (Beta-Blocker)

C) Ibuprofen

D) Codeine

> 💡 **Hint:** Beta-blockers like Propranolol allow unopposed alpha-adrenergic activity, keeping the cranial vessels slightly constricted to prevent the painful vasodilation phase.

58. Verapamil is sometimes used for migraine prophylaxis. It belongs to which class?

A) Beta-blocker

B) Calcium Channel Blocker

C) Antidepressant

D) Anticonvulsant

> 💡 **Hint:** Calcium channel blockers reduce the frequency of migraine attacks, though they may not affect the severity or duration.

59. Full Agonist opioids (like Morphine) bind to receptors and produce:

A) Sub-maximal effect

B) Maximal effect with high addiction liability

C) No effect

D) Antagonistic effect

> 💡 **Hint:** Strong agonists have high efficacy and high abuse/addiction liability due to profound euphoria.

60. Which of the following is classified as a Partial Agonist opioid?

A) Morphine
B) Naloxone
C) Buprenorphine
D) Fentanyl

> 💡 **Hint:** Buprenorphine is a partial agonist at the Mu receptor. It acts like morphine in naive patients but can precipitate withdrawal in dependent users.

61. Pure Antagonist opioids (like Naloxone) possess which pharmacological properties?

A) High affinity and high efficacy
B) High affinity but zero efficacy
C) Low affinity and low efficacy
D) No affinity

> 💡 **Hint:** Antagonists bind strongly to the opioid receptors (high affinity) but do not activate them (zero efficacy), displacing agonists like heroin.

62. The primary analgesic, euphoric, and respiratory-depressant effects of opioids are mediated by which receptor?

A) Kappa (κ)
B) Delta (δ)
C) Mu (μ)
D) Sigma (σ)

> 💡 **Hint:** The Mu (μ) receptor is the most clinically important opioid receptor, responsible for both the profound analgesia and the dangerous respiratory depression.

63. Opioids inhibit pain transmission presynaptically by decreasing the influx of which ion?

A) Potassium (K+)

B) Sodium (Na+)

C) Calcium (Ca2+)

D) Chloride (Cl-)

> 💡 **Hint:** Blocking presynaptic Calcium influx prevents the synaptic vesicles from releasing excitatory neurotransmitters like glutamate.

64. Postsynaptically, opioids cause neuronal hyperpolarization by increasing the efflux of which ion?

A) Calcium (Ca2+)

B) Potassium (K+)

C) Magnesium (Mg2+)

D) Sodium (Na+)

> 💡 **Hint:** Opening Potassium channels lets K+ leave the cell, making the inside more negative (hyperpolarized), thus inhibiting nerve firing.

65. Morphine relieves severe pain without causing:

A) Euphoria

B) Loss of consciousness

C) Constipation

D) Miosis

> 💡 **Hint:** Morphine provides profound analgesia; the patient is still aware of the pain, but it is no longer unpleasant, and they do not lose consciousness.

66. The most common cause of death in acute Opioid overdose is:

A) Constipation and bowel rupture

B) Respiratory Depression

C) Pruritus and anaphylaxis

D) Urinary Retention

> 💡 **Hint:** Respiratory depression is the fatal side effect. It occurs because opioids reduce the sensitivity of the brainstem's respiratory center to CO_2.

67. Pinpoint pupils (Miosis) in a comatose patient are a diagnostic sign of:

A) Cocaine overdose

B) Opioid overdose

C) Atropine poisoning

D) Alcohol intoxication

> 💡 **Hint:** Most causes of coma dilate the pupils. Opioid overdose uniquely causes pinpoint pupils. Crucially, NO tolerance develops to this miotic effect.

68. The following are pharmacological actions of morphine EXCEPT: (Midterm Question)

A) Miosis

B) Physical dependence

C) Euphoria

D) Diarrhea

> 💡 **Hint:** Morphine causes severe CONSTIPATION by decreasing motility and increasing sphincter tone, not diarrhea.

69. Morphine can worsen pain in patients suffering from biliary colic because it:

A) Relaxes the gallbladder

B) Causes Biliary Spasm (constricts the sphincter of Oddi)

C) Decreases bile production

D) Increases GI motility

> 💡 **Hint:** Morphine increases biliary tract pressure significantly, making it unsuitable for biliary colic pain.

70. Morphine is contraindicated in patients with severe Bronchial Asthma because it causes:

A) Beta-receptor blockade

B) Histamine release from mast cells

C) Leukotriene synthesis

D) Respiratory stimulation

> 💡 **Hint:** Morphine triggers non-immunologic histamine release, which leads to vasodilation, sweating, and dangerous bronchoconstriction in asthmatics.

71. Morphine is not recommended for analgesia during the active stage of childbirth because it:

A) Causes fetal malformations

B) Prolongs labor by decreasing uterine contractions

C) Causes maternal hemorrhage

D) Stimulates premature pushing

> 💡 **Hint:** Morphine transiently decreases the strength, duration, and frequency of uterine contractions, prolonging the second stage of labor. Meperidine is preferred here.

72. Morphine produces vomiting due to direct stimulation of the chemoreceptor trigger zone (CTZ). (Midterm Question)

A) True

B) False

💡 **Hint:** True. Morphine stimulates the CTZ in the area postrema of the medulla, leading to nausea and emesis, especially in ambulatory patients.

73. Fentanyl is a synthetic opioid that is approximately how many times more potent than Morphine?

A) 10x

B) 50x

C) 100x

D) 500x

💡 **Hint:** Fentanyl has 100-fold the analgesic potency of morphine. It is highly lipophilic with a very rapid onset and short duration.

74. Fentanyl is heavily preferred during cardiac surgery anesthesia because it provides:

A) Stable hemodynamics with negligible effects on myocardial contractility

B) Profound hypotension to reduce bleeding

C) Tachycardia to maintain cardiac output

D) Bronchodilation

💡 **Hint:** Unlike morphine which can cause hypotension via histamine, fentanyl has minimal cardiovascular effects, keeping hemodynamics perfectly stable.

75. Meperidine (Pethidine) has a toxic metabolite called Normeperidine which can accumulate and cause:

A) Severe respiratory depression

B) Seizures, tremors, and muscle twitches

C) Chronic constipation

D) Fatal Miosis

💡 **Hint:** Large or repetitive doses of Meperidine lead to normeperidine accumulation, which causes dangerous CNS excitation and seizures.

76. Unlike Morphine, high doses of Meperidine uniquely cause which effect on the eye?

A) Pinpoint pupils

B) Mydriasis (Pupil dilation)

C) Nystagmus

D) Cataracts

💡 **Hint:** Meperidine has significant antimuscarinic (atropine-like) activity, which dilates the pupil instead of constricting it.

77. Meperidine is absolutely contraindicated in patients taking MAO Inhibitors due to the risk of:

A) Hypotensive crisis

B) Hyperpyrexic Coma (Serotonin Syndrome)

C) Acute renal failure

D) Hepatic necrosis

💡 **Hint:** The combination of Meperidine and MAOIs can provoke severe, fatal reactions including convulsions and hyperthermia.

78. Methadone is heavily utilized clinically for the controlled withdrawal and detoxification of heroin addicts because:

A) It is a pure antagonist

B) It is short-acting and rapidly cleared

C) It induces a milder, but more protracted withdrawal syndrome due to its long duration of action

D) It lacks any abuse liability

💡 **Hint:** Methadone is a strong Mu-agonist but has a very long half-life. It substitutes for heroin, preventing cravings, and allows for a smooth, slow weaning process.

79. Codeine is primarily used clinically as a moderate analgesic and as an:

A) Anti-emetic

B) Antitussive (Cough suppressant)

C) Anti-diarrheal

D) Anesthetic adjunct

💡 **Hint:** Codeine has strong antitussive properties at sub-analgesic doses.

80. To exert its analgesic effect, Codeine must be enzymatically converted in the body into:

A) Heroin

B) Morphine

C) Norcodeine

D) Thebaine

💡 **Hint:** Codeine itself is a prodrug for analgesia; it relies on CYP2D6 conversion to Morphine to relieve pain.

81. ________ is a relatively safe non-addicting antitussive agent that replaced codeine in most nonprescription cough syrups. (Midterm Question)

A) Morphine

B) Methadone

C) Meperidine

D) Dextromethorphan

> 💡 **Hint:** Dextromethorphan is a synthetic cough depressant with virtually no analgesic action and low abuse potential.

82. Tramadol is classified as an atypical analgesic because, in addition to weak Mu-receptor binding, it inhibits the reuptake of:

A) Dopamine and GABA

B) Norepinephrine and Serotonin

C) Acetylcholine and Histamine

D) Glutamate and Glycine

> 💡 **Hint:** Tramadol acts centrally. It weakly binds Mu receptors but also boosts NE and Serotonin levels to modulate pain pathways.

83. All of the following drugs are opioid mixed agonist-antagonists EXCEPT: (Midterm Question)

A) Naloxone

B) Butorphanol

C) Nalbuphine

D) Buprenorphine

> 💡 **Hint:** Naloxone is a PURE competitive antagonist, not a mixed agonist-antagonist.

84. If a partial agonist/mixed agonist (like Pentazocine) is given to a patient physically dependent on full agonists (like Morphine), it will:

A) Enhance the euphoria

B) Precipitate immediate Withdrawal Symptoms

C) Have no clinical effect

D) Cause fatal respiratory depression

> 💡 **Hint:** Because they have high affinity but lower efficacy, they competitively kick the full agonist off the receptor, acting effectively as an antagonist in dependent patients.

85. The opioid mixed agonist/antagonist may act as an opioid antagonist in: (Midterm Question)

A) Buprenorphine-treated patients

B) Morphine-dependent patients

C) Nalbuphine-treated patients

D) Naloxone-treated patients

> 💡 **Hint:** As explained by the Doctor, giving a mixed agent to a Morphine-addict displaces the morphine, acting as a functional antagonist and triggering withdrawal.

86. Buprenorphine is highly effective for opiate detoxification. It is typically administered via which route?

A) Intravenous

B) Sublingual

C) Intramuscular

D) Transdermal patch only

> 💡 **Hint:** Buprenorphine tablets are administered sublingually for opioid dependence because it binds tightly to Mu receptors with a long duration of action.

87. Opioid pure antagonists include: (Midterm Question)

A) Naloxone
B) Naltrexone
C) Nalbuphine
D) A and B

> 💡 **Hint:** Both Naloxone and Naltrexone are pure competitive antagonists at opioid receptors.

88. Which drug is used clinically for the emergency reversal of an Opioid Overdose coma? (Midterm Question)

A) Naltrexone
B) Naloxone
C) Dextromethorphan
D) Codeine

> 💡 **Hint:** Naloxone given IV acts within 30 seconds to dramatically reverse the coma and respiratory depression of heroin/morphine overdose.

89. A critical clinical warning regarding Naloxone administration in overdose patients is:

A) It causes permanent receptor damage
B) Its duration of action (half-life) is shorter than most opioids, risking relapse into a coma
C) It causes immediate cardiac arrest
D) It only works on synthetic opioids

> 💡 **Hint:** Naloxone clears quickly (1-2 hours). If the patient overdosed on a long-acting opioid, they will stop breathing again once the Naloxone wears off.

90. Naltrexone differs from Naloxone primarily because Naltrexone:

A) Is used for acute overdose emergencies

B) Has a much longer duration of action (up to 48 hours) and is orally active

C) Is a partial agonist

D) Causes severe respiratory depression

> 💡 **Hint:** Naltrexone is given orally as maintenance therapy for detoxified addicts to block the effects of any illicit heroin they might inject.

91. Tolerance develops to almost all effects of Morphine with repeated use, EXCEPT:

A) Analgesia and Euphoria

B) Respiratory depression

C) Miosis (pinpoint pupil) and Constipation

D) Sedation and Nausea

> 💡 **Hint:** The Doctor stressed this: An addict taking massive doses of heroin will still have pinpoint pupils and chronic constipation. Tolerance NEVER develops to these.

92. Physical dependence on opioids is clinically characterized by:

A) Psychological craving for euphoria only

B) The onset of a severe Withdrawal Syndrome upon abrupt cessation or antagonist administration

C) Permanent brain damage

D) Increased tolerance to constipation

> 💡 **Hint:** Physical dependence means the body requires the drug to function normally. Removing the drug triggers horrific physical withdrawal symptoms.

93. Symptoms of Opioid Withdrawal include all of the following EXCEPT:

A) Anxiety and severe muscle/joint pain

B) Dilated Pupils (Mydriasis)

C) Diarrhea and Vomiting

D) Miosis (Pinpoint pupils)

> 💡 **Hint:** Withdrawal symptoms are generally the OPPOSITE of the drug's effects. Since opioids cause miosis and constipation, withdrawal causes mydriasis and diarrhea.

94. According to the Doctor's lecture, what becomes the primary 'driving force' that compels an addict to continue seeking opioids in the later stages of addiction?

A) The intense initial euphoria

B) Pain management

C) The absolute terror of experiencing the severe Withdrawal Syndrome

D) Peer pressure

> 💡 **Hint:** The doctor explained that while euphoria starts the addiction, avoiding the excruciating pain and symptoms of the withdrawal syndrome is what keeps the addict trapped.

95. Combining Opioids with other CNS depressants like Benzodiazepines or Alcohol is extremely dangerous because it can lead to:

A) Enhanced euphoria without risks

B) Synergistic and fatal Respiratory Depression

C) Increased alertness and seizures

D) Immediate tolerance reversal

> 💡 **Hint:** Both classes depress the respiratory center. Taking them together creates a deadly synergy that easily causes the patient to stop breathing.

96. The eye pupil of a morphine addict is a diagnostic tool for: (Midterm Question)

A) Opioid overdose coma

B) Opioid addiction

C) Hypertension

D) A and B

> 💡 **Hint:** Pinpoint pupils (Miosis) allow ER doctors to immediately diagnose an overdose coma, and police to identify addicts, because no tolerance develops to this effect.

97. ________ is NOT an opioid antagonist. (Midterm Question)

A) Naloxone

B) Nalbuphine

C) Naltrexone

D) Nalmefene

> 💡 **Hint:** Nalbuphine is a mixed agonist-antagonist used for pain, not a pure antagonist like Naloxone, Naltrexone, and Nalmefene.

98. Which of the following describes the mechanism by which opioids cause euphoria?

A) Direct stimulation of the cerebral cortex

B) Disinhibition of the ventral tegmentum leading to dopamine release

C) Blockade of serotonin reuptake

D) Stimulation of the Chemoreceptor Trigger Zone

> 💡 **Hint:** Opioids produce a powerful sense of well-being (euphoria) by disinhibiting the ventral tegmentum, a key part of the brain's reward circuit.

99. Which opioid has strong antimuscarinic effects, is shorter acting than morphine, and is widely used in obstetrics?

A) Fentanyl

B) Methadone

C) Meperidine

D) Codeine

> 💡 **Hint:** Meperidine is preferred in labor because it causes less prolongation of uterine contractions compared to morphine, and has distinct atropine-like effects.

100. Detoxification of a heroin-dependent individual is usually accomplished through the slow, oral administration of:

A) Naloxone

B) Fentanyl

C) Methadone or Buprenorphine

D) Meperidine

> 💡 **Hint:** Methadone (long-acting full agonist) or Buprenorphine (long-acting partial agonist) substitute for heroin, preventing cravings while allowing for a slow, tolerable dose reduction.